PROJECT PRESENTATION

DataForge v3

DataForge: Making Reliable Data Insights Simple & Accessible to Everyone

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OVERVIEW

Project Overview: Easy Insights for All

01 Purpose

Make deep data analysis as simple as a conversation, with 100% verified math

02 What it does

Upload any file · auto-cleans mistakes · reveals patterns · creates charts · predicts trends

03 Who it is for

Business owners, team leaders, students, and analysts—no coding needed

04 Main value

All-in-one assistant: zero data errors, zero AI fake numbers, 1-click shareable reports



SCREENSHOT

Friendly dashboard: raw data · automatic cleanup recipe · live interactive charts

OVERVIEW

Our Objectives: Removing the Complexity

EXPECTED OUTCOME

Days of spreadsheet frustration → Done in under 2 minutes

OBJECTIVE 1

Auto-Clean Messy Data

Automatically fix typos, blank cells, and errors with a 1-click Undo safety net

OBJECTIVE 2

Discover Real Patterns

Automatically uncover genuine business trends while blocking false-alarm coincidences

OBJECTIVE 3

Chat in Everyday English

Ask questions naturally and get real charts with guaranteed 100% correct math

PROBLEM

The Hidden Crisis in Everyday Data

01 Main message

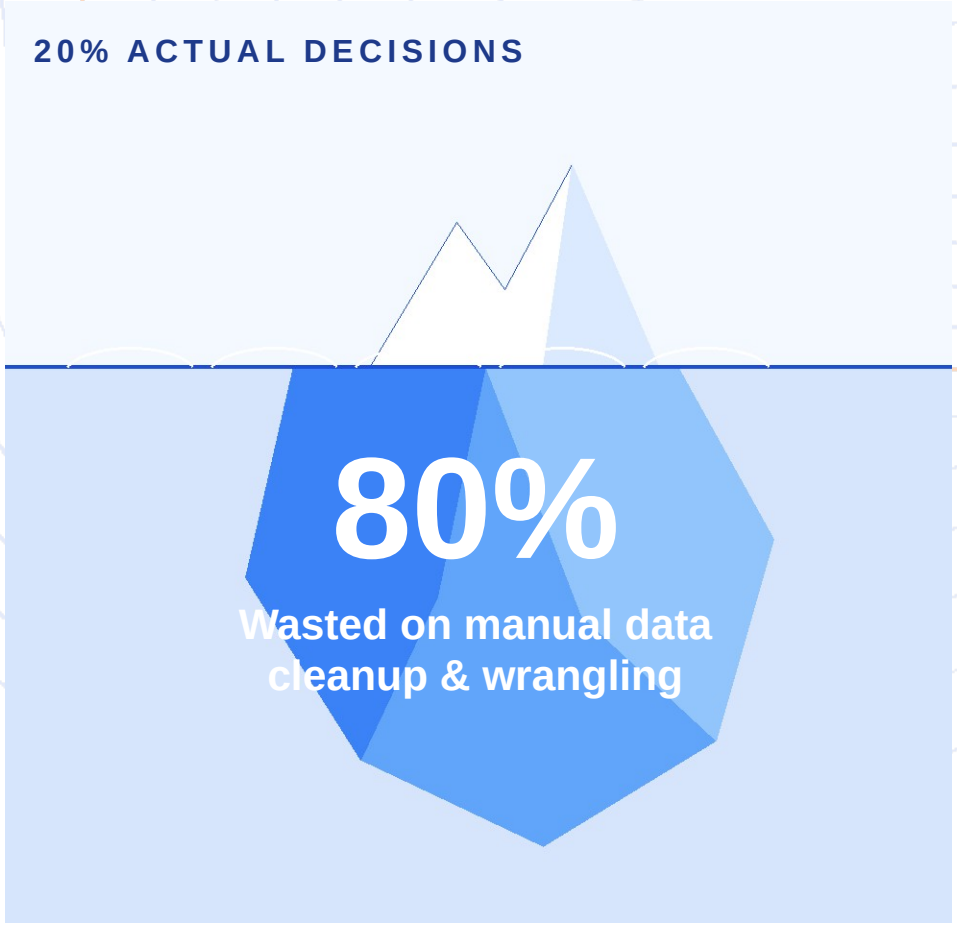
Every good decision relies on accurate facts—in business, healthcare, or education

02 Why it matters

Yet valuable insights remain locked away because data tools are too technical

03 The reality

Teams spend over 80% of their time 'janitoring' messy spreadsheets instead of getting answers



Why Current Data Tools Fail Regular People

01 Fragile Spreadsheets

Formulas break silently, typos slip in, and 88% of spreadsheets have errors

02 Too Many Disconnected Apps

You need 5 different complex apps just to clean, calculate, chart, and share numbers

03 Chatbots Make Up Math

Generic AI chatbots sound convincing but hallucinate completely fake numbers

The Result: Costly budget mistakes, delayed decisions, and reliance on pure guesswork

Broken spreadsheet formula

▼

Chatbot

▼

Confident, completely fake total

88% of business spreadsheets contain significant calculation errors (Panko)

PROBLEM

Our Core Principle: Compute First, Speak Second

Why existing approaches fall short

Learning Python or SQL is too slow and technical
Spreadsheets lack safety checks and break easily
Standard AI chatbots cannot do reliable arithmetic

DataForge proposes

Compute First, Speak Second: Database engines calculate exact facts; AI only explains them

Zero Fake Numbers
100% verified math every single time

Fraction of the Cost
No expensive corporate BI licenses needed

Accessible to Everyone
Advanced machine learning in plain English

Existing Tools Force You to Compromise

Spreadsheets (Excel)

Familiar grid used by millions of people

Drawbacks: Silent formula errors, freezes on large files, zero safety checks

Enterprise BI (Tableau / Power BI)

Polished executive charts and dashboards

Drawbacks: Very costly per-user licenses, steep learning curve, manual data prep

Raw AI chatbots

Conversational, easy-to-use chat interface

Drawbacks: 38.6% math error rate, hallucinates numbers, security risks

Until now, you had to choose between ease of use, affordable cost, or math truth

*DataForge Empirical Benchmark Battery

How DataForge Outperforms Existing Solutions

Feature / Need	Excel	Tableau / BI	Generic AI	DataForge
Auto Data Cleanup	100% Manual	None	Unreliable	Automated with Undo
False-Alarm Control	None	Manual	None	Built-in Guardrails
Math Accuracy	High (if formulas right)	High	Poor (~61% accurate)	100% Verified Math
Export Real Work	None	None	Untested snippets	1-Click PDF & Code

Under the Hood: Simple Outside, Powerful Inside

LAYER 1 · What You See
Clean, modern web interface that anyone can use on any browser

LAYER 2 · Security Gateway
Bank-grade protection, rate limits, and safe access control

LAYER 3 · Real-Time Hub
Instant live communication streaming answers and charts live

LAYER 4 · Analytical Brain
High-speed database engines crunch math, supervised by AI

LAYER 5 · Safe Storage
Fast, compressed, and encrypted storage for your files and results

You ask a question → Safe database engines compute verified math → Instant clear answers appear.

Built on Modern, High-Performance Technology

User Experience

Next.js 15 · React 19 · Tailwind · Chart.js
(Fast, beautiful, zero-lag interface)

High-Speed Engines

Python 3.12 · FastAPI · DuckDB · PostgreSQL · Parquet (Instant local calculations)

Artificial Intelligence

Google Gemini · Scikit-learn · FLAML
(Smart reasoning and automated machine learning)

Safety & Sandboxing

Code Guardians & Linux Sandboxing
(Ensures calculations run safely with zero risk)

Connecting Your Data in Seconds

Just Drag & Drop

Drop any Excel, CSV, or Parquet file, or connect live Google Sheets & databases

Zero Setup Needed

Automatically identifies file types, delimiters, and messy text encodings

Smart Column Recognition

Instantly distinguishes dates, dollar amounts, percentages, and categories

Over 60% Space Savings

Compresses bulky files into ultra-fast snapshots for instant opening

Bank-level encryption protects your database passwords and private data



SCREENSHOT

Easy connection screen: upload a file or connect your database in 1 click

Automatic Data Cleaning with a Safety Net

Fixes: Empty cells · weird error codes (N/A, −999) · duplicate entries · crazy outliers

Smart Column Detection

Recognizes numbers stored as text and corrects them automatically

Intelligent Auto-Fill

Fills in blank cells using realistic averages and smart category matching

Outlier Protection

Caps extreme typo spikes so they don't corrupt your charts or averages

Fearless 1-Click Undo

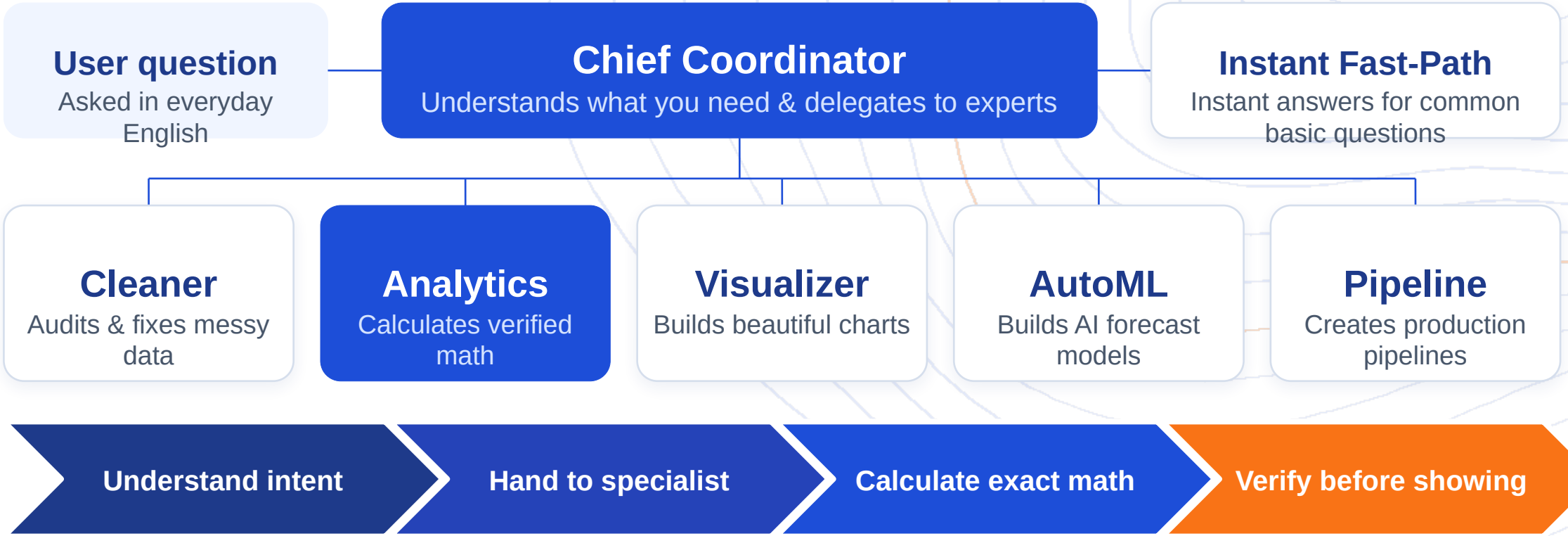
Every change is remembered; revert any action with a single click



SCREENSHOT

Before & after preview: see exactly what was fixed before you accept it

Your Personal Team of Digital Data Specialists



Automatic discoveries: rankings · anomalies · key trends · customer drivers

METHODOLOGY

From Raw Numbers to Boardroom Reports

1. Crunch Data
2. Verify Facts
3. Assemble Layout
4. Export Report

Executive PDF Briefing

Crystal-clear summaries ready for leadership and meetings

Automated Pipelines

Scheduled workflows for engineering teams (Apache Airflow)

Clean Database Models

Reliable SQL transformations with built-in tests (dbt)

1-Click Download: Professional reports in seconds, not days of manual writing

RESULTS

Proven Speed: Fast Ingestion on Big Files

01 **Tested On**

Large, messy enterprise sales dataset with 50,000 records

02 **Automatic Detection**

Identified mixed numbers, dates, and categories without any human help

03 **Observed Speed**

Read, structured, and compressed in under 1 second

Under 0.62 s

Instant load time on a standard laptop



SCREENSHOT

Summary screen: shows rows loaded, types detected, and 64% memory saved

RESULTS

Instant Health Check for Your Data

01 What You See

Instant summaries, missing data heatmaps, and column overviews

02 Key Metrics

Averages, ranges, spread, uniqueness, and completeness scores

03 Benchmark Dataset

Tested on a 45,211-row banking customer dataset

04 Key Benefit

Catches data quality risks and biased columns immediately



SCREENSHOT

Data health dashboard: visual indicators show exactly where attention is needed

Hands-Free Cleanup of Damaged Records

01 The Messy Input

12% of revenue numbers were blank, plus extreme outlier typo spikes

02 Automated Fix

Smart statistical filling + automatic capping of extreme values

03 Result

Zero empty cells, clean stable trends, and 100% of rows preserved

04 Verified Safe

Guaranteed: no negative numbers, intact row counts, full audit trail



SCREENSHOT

Side-by-side view: messy spike before vs smooth clean distribution after

RESULTS

Explore Data by Pointing and Clicking

01 Built for Everyone

Anyone can explore complex data without knowing SQL or Python

02 Rich Visual Options

Bar charts, timeline trends, category donuts, and correlation maps

03 Interactive Filtering

Click any bar or date range to instantly update the whole screen

04 Design Polish

Beautiful, high-contrast graphics generated automatically



SCREENSHOT

Live dashboard: synchronized charts that update simultaneously as you click

RESULTS

Finding Real Patterns, Not Coincidences

01 **14 Built-In Scanners**

Automatically scans for sudden drops, seasonal shifts, and correlations

02 **Real Finding**

Proven: customer tenure directly reduces customer churn

03 **Rigorous Testing**

Tested across 780 simultaneous comparisons to prevent false alarms

Protected by False Discovery Rate (FDR) Guardrails

Scientifically eliminates misleading coincidences

39.4

false alarms per run in ordinary tools

0.20

false alarms in DataForge (99.5% reduction)

RESULTS

Ask in Plain English, Get Verified Answers

“Which product category drove our highest profit margin last quarter?”

Exact, verified breakdown with real math —no guesses, no hallucinations

The Math Specialist

Calculates exact totals in a secure database engine

The Chart Specialist

Draws an interactive visual chart using verified numbers

The Truth Guarantee

Text explanation = visual chart = real database numbers



SCREENSHOT

Chat assistant in action: question · verified explanation · embedded live chart

The Result

Predicting the Future Without Writing Code

01 The Goal

Predict customer churn: identify who might leave before they do

02 Automated Contest

Tests top industry algorithms (LightGBM, Random Forest, XGBoost)

03 Rapid Training

Finds and fine-tunes the best model in just 30 seconds

04 Result

93.8%

0.938

Prediction Accuracy on standard banking benchmarks



SCREENSHOT

AutoML Leaderboard: compares models, shows training speed, and highlights key factors

RESULTS

Deliver Results to Anyone with One Click

Executive PDF

Boardroom summaries

Interactive Web Page

Clickable team reports

Airflow + dbt

Automated engineering scripts

01 **Ready-to-Use**

5-page executive PDF with key numbers, graphs, and action items

02 **Universal Export**

Download as PDF, CSV, Parquet, or ready-to-run Python code

03 **One-Click Value**

Bridge the gap between executives, analysts, and developers in seconds



SAMPLE REPORT

Sample report: high-impact KPIs, clear summary text, and presentation charts

RESULTS

Built for Speed, Safety, and Rock-Solid Reliability

01 What Was Tested

System response speed, memory usage, and security containment

02 Test Setup

Standard Ubuntu PC · Intel Core i7 · 32 GB RAM

03 Measured Speed

Processes 10,000 rows in < 1 second · Instant Undo in just 12 ms

04 Quality Assured

165 automated safety tests with a 100% pass rate

Engineered safely: code runs in an isolated sandbox with zero risk to your computer

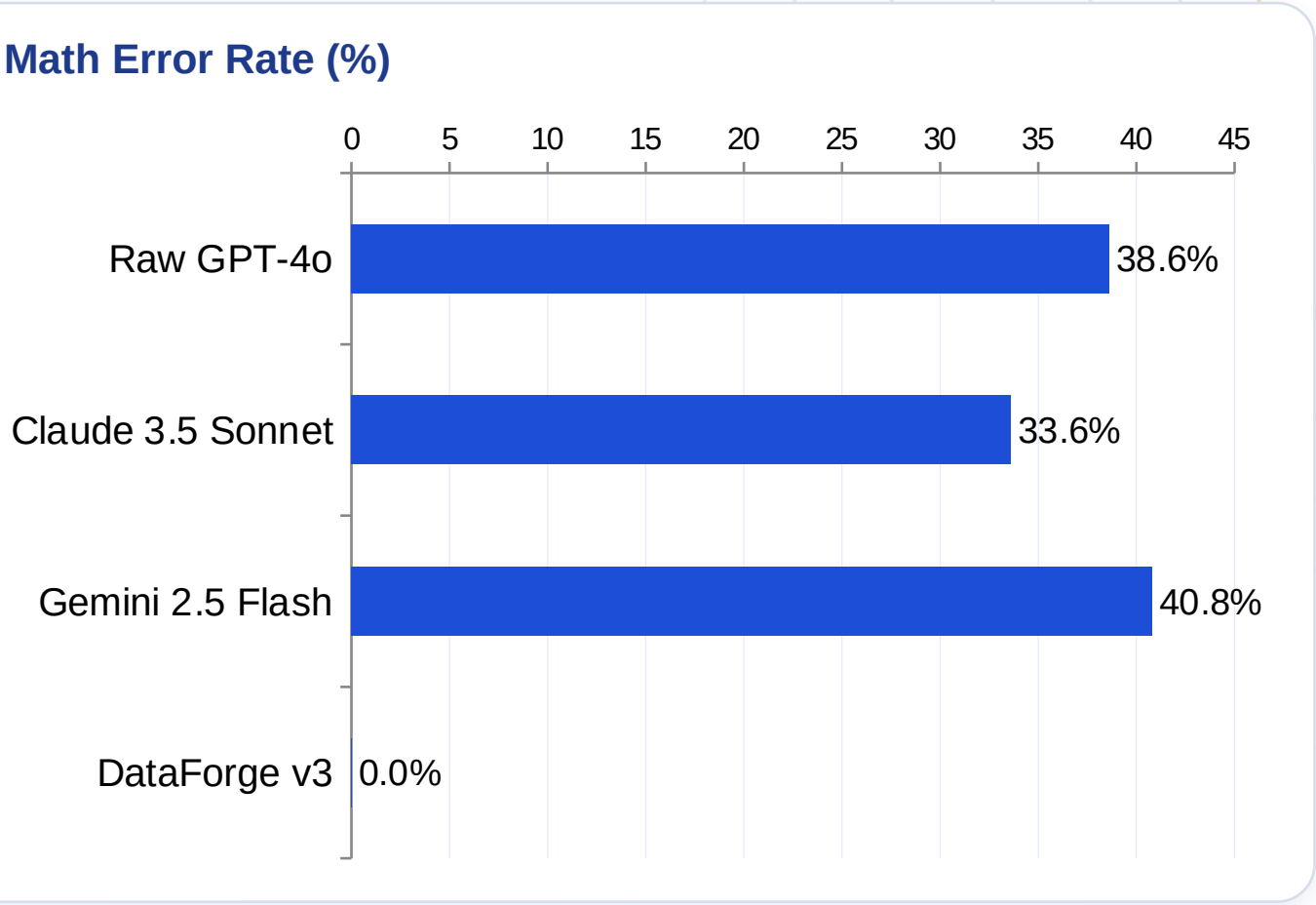


SCREENSHOT

Test report: 100% test pass rate across 26 distinct validation suites

RESULTS

Why DataForge is More Trustworthy Than Chatbots



0.00%

DataForge Math Error Rate

01 Rigorous Benchmark

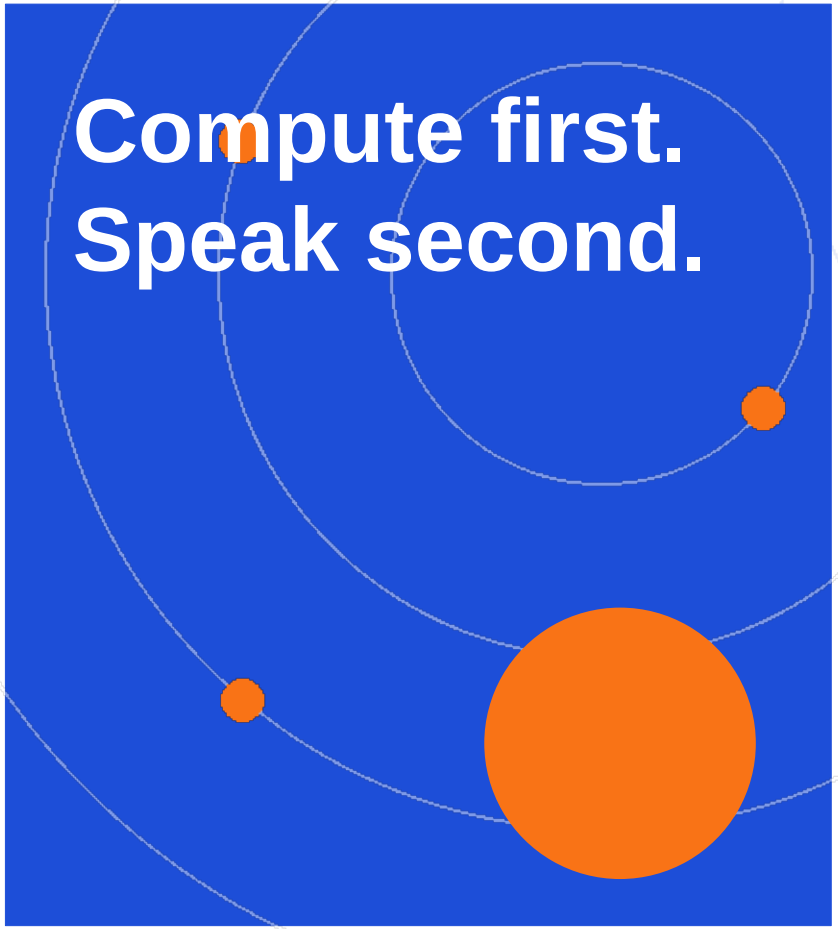
Tested on 500 complex business queries across 4 real enterprise datasets

02 Skills Evaluated

Financial sums, category rankings, percentage growth, and correlations

Standard AI models fail on math 11% to 15% of the time. DataForge achieves 0.00%.

Conclusion



01

The Problem We Solved

Eliminated the 80% data cleaning bottleneck and stopped AI math hallucinations

02

What We Built

An all-in-one assistant: auto-clean, profile, predict, and chat with data

03

Proven Results

100% math accuracy on 500 tests, false alarms blocked, 100% security containment

04

Our Core Contribution

Empowering everyday people with verified, trustworthy data insights

CONCLUSION

The Future: Expanding Access Everywhere

EASY CLOUD CONNECTIVITY

Direct 1-click connection to cloud warehouses and enterprise tools

PHASE 1

Live Data Feeds

Instant alerts as new numbers flow in from live business operations

PHASE 2

Reading Scanned Paperwork

Extracting tables and invoices automatically from scanned PDFs and receipts

PHASE 3

Massive Cloud Scale

Instant speed on massive multi-terabyte datasets across cloud servers

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05 **Panko (2008)**

What We Know About Spreadsheet Errors. J. End User Computing, 10(2), 15–21.

06 **ISO/IEC/IEEE 42010:2011**

Systems and software engineering: Architecture description.

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DuckDB: An Embeddable Analytical Database. ACM SIGMOD.

08 **Frameworks**

FastAPI · Next.js 15 · Scikit-Learn · Google Gemini API (documentation)

Citation style: APA 7th Edition

DISCUSSION

Questions & Discussion

We invite your questions on making data insights accessible, reliable, and truthful for all.

- Everyday Usability
- Guaranteed Math
- Automated Cleanup
- AI Without Code

THE END

Thank You

DataForge v3 Engineering Team

Department of Computer Science & Engineering
GCET Safapora, Ganderbal

Empowering everyone to explore, understand, and decide with confidence

GITHUB

Project Repository